Ama Febriyanti/ Universitas Negeri Malang

Topic 15, 16, 17 - Assignment | Hands-On SQL

1. Identify the top 10 customers and their email so we can reward them

   sql query:

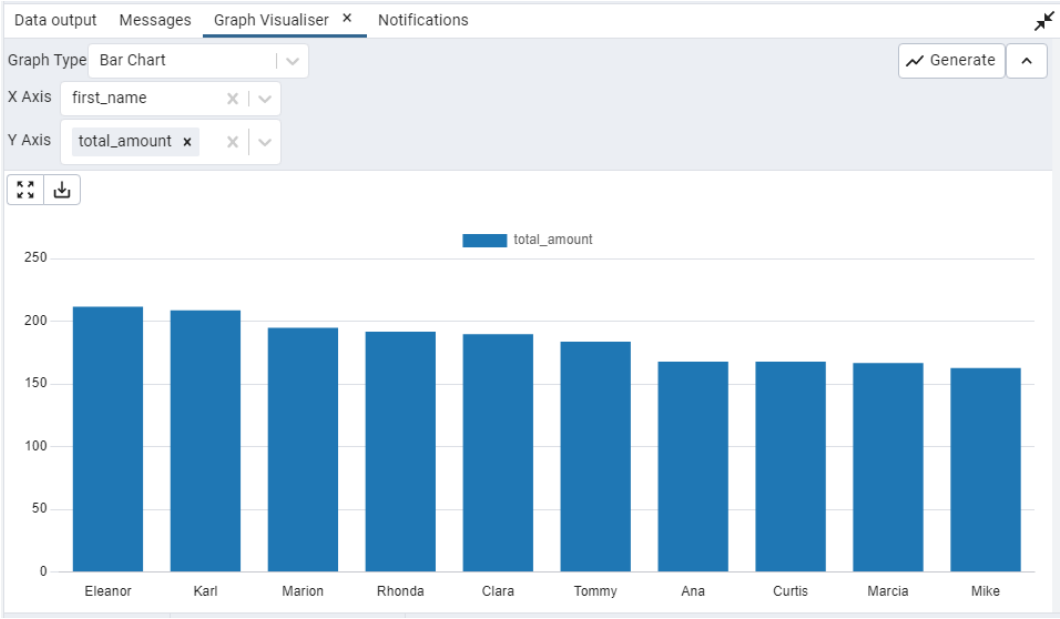
   ```
   select distinct a.customer_id, first_name, last_name, email, sum(amount) total_amount
   from customer a
   join payment b
   on a.customer_id=b.customer_id
   group by a.customer_id
   order by sum(amount) desc
   limit 10
   ```

   screenshot of query result:

Data output | Messages | Notifications

	customer_id [PK] integer	first_name character varying (45)	last_name character varying (45)	email character varying (50)	total_amount numeric
1	148	Eleanor	Hunt	eleanor.hunt@sakilac...	211.55
2	526	Karl	Seal	karl.seal@sakilacusto...	208.58
3	178	Marion	Snyder	marion.snyder@sakil...	194.61
4	137	Rhonda	Kennedy	rhonda.kennedy@saki...	191.62
5	144	Clara	Shaw	clara.shaw@sakilacu...	189.60
6	459	Tommy	Collazo	tommy.collazo@sakil...	183.63
7	181	Ana	Bradley	ana.bradley@sakilacu...	167.67
8	410	Curtis	Irby	curtis.irby@sakilacus...	167.62
9	236	Marcia	Dean	marcia.dean@sakilac...	166.61
10	403	Mike	Way	mike.way@sakilacust...	162.67

   result visualization:



2. Identify the bottom 10 customers and their emails

sql query:

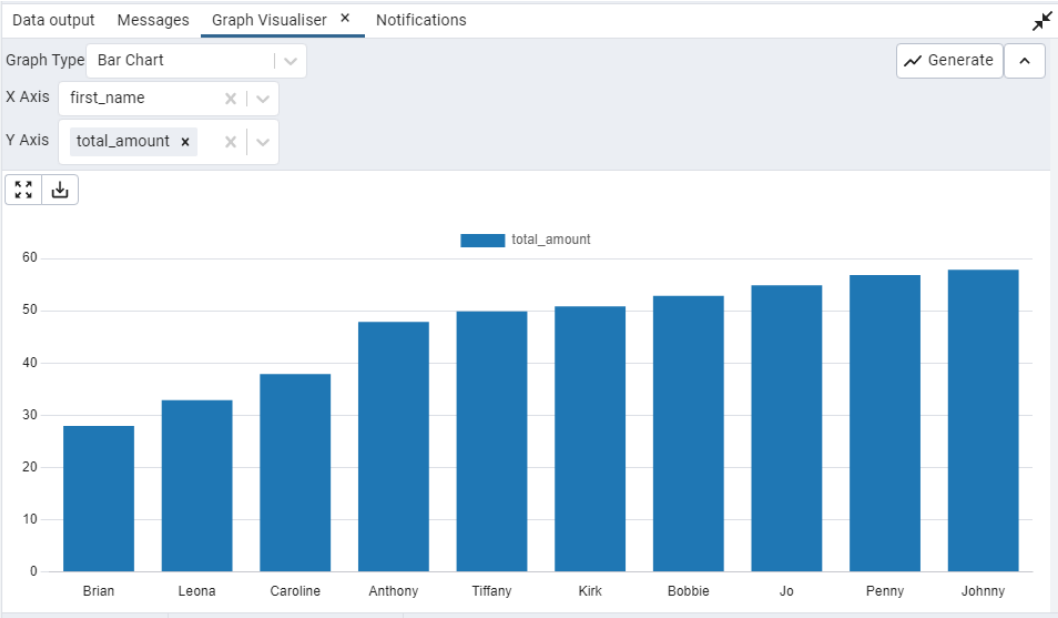
```
select distinct a.customer_id, first_name, last_name, email, sum(amount) total_amount
from customer a
join payment b
on a.customer_id=b.customer_id
group by a.customer_id
order by sum(amount)
limit 10
```

screenshot of query result:

Data output | Messages | Notifications

	customer_id [PK] integer	first_name character varying (45)	last_name character varying (45)	email character varying (50)	total_amount numeric
1	318	Brian	Wyman	brian.wyman@sakilacustomer.org	27.93
2	281	Leona	Obrien	leona.obrien@sakilacustomer.org	32.90
3	248	Caroline	Bowman	caroline.bowman@sakilacustomer...	37.87
4	320	Anthony	Schwab	anthony.schwab@sakilacustomer....	47.85
5	110	Tiffany	Jordan	tiffany.jordan@sakilacustomer.org	49.88
6	586	Kirk	Stclair	kirk.stclair@sakilacustomer.org	50.83
7	288	Bobbie	Craig	bobbie.craig@sakilacustomer.org	52.81
8	250	Jo	Fowler	jo.fowler@sakilacustomer.org	54.85
9	271	Penny	Neal	penny.neal@sakilacustomer.org	56.84
10	395	Johnny	Turpin	johnny.turpin@sakilacustomer.org	57.81

result visualization:



3. What are the most profitable movie genres (ratings)?

sql query:

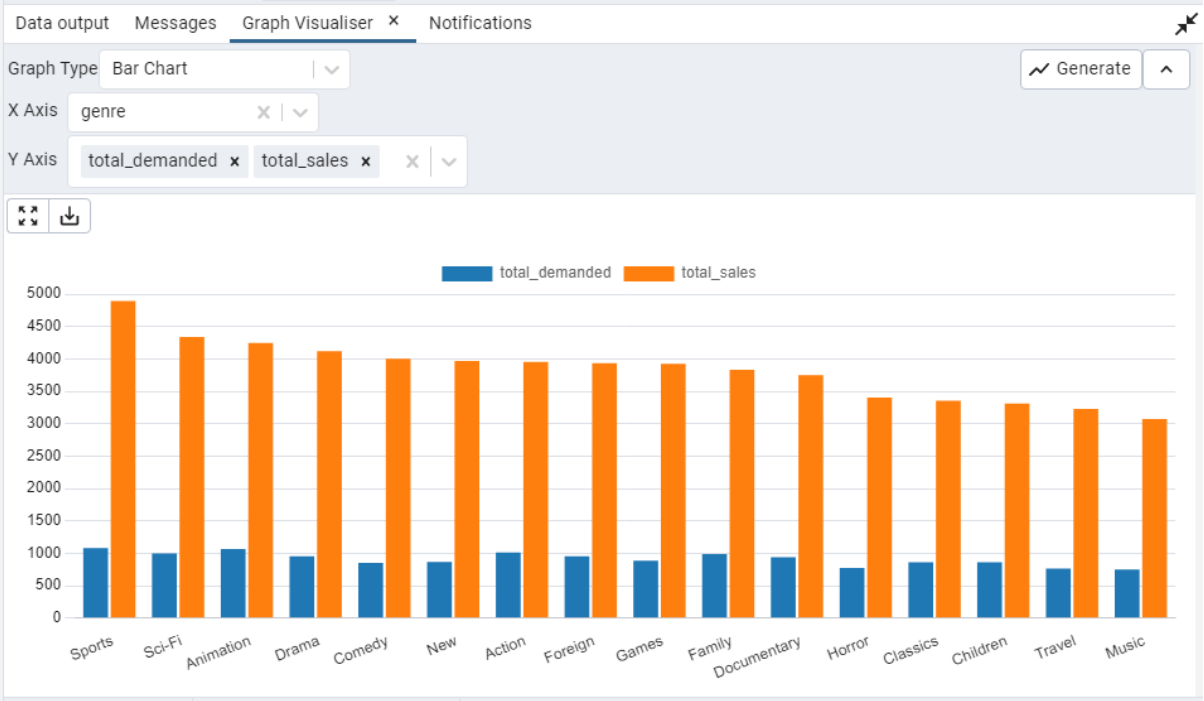
```
select distinct a.name genre, count(f.customer_id) total_demanded, sum(g.amount)
total_sales
from category a
join film_category b on a.category_id=b.category_id
join film c on b.film_id=c.film_id
join inventory d on c.film_id=d.film_id
join rental e on d.inventory_id=e.inventory_id
join customer f on e.customer_id=f.customer_id
join payment g on e.rental_id=g.rental_id
group by a.name
order by sum(g.amount) desc
limit 1
```

screenshot of query result:

Data output Messages Notifications

	genre character varying (25)	total_demanded bigint	total_sales numeric
1	Sports	1081	4892.19

result visualization (without limit):



4. How many rented movies were returned late, early, and on time?
   sql query:

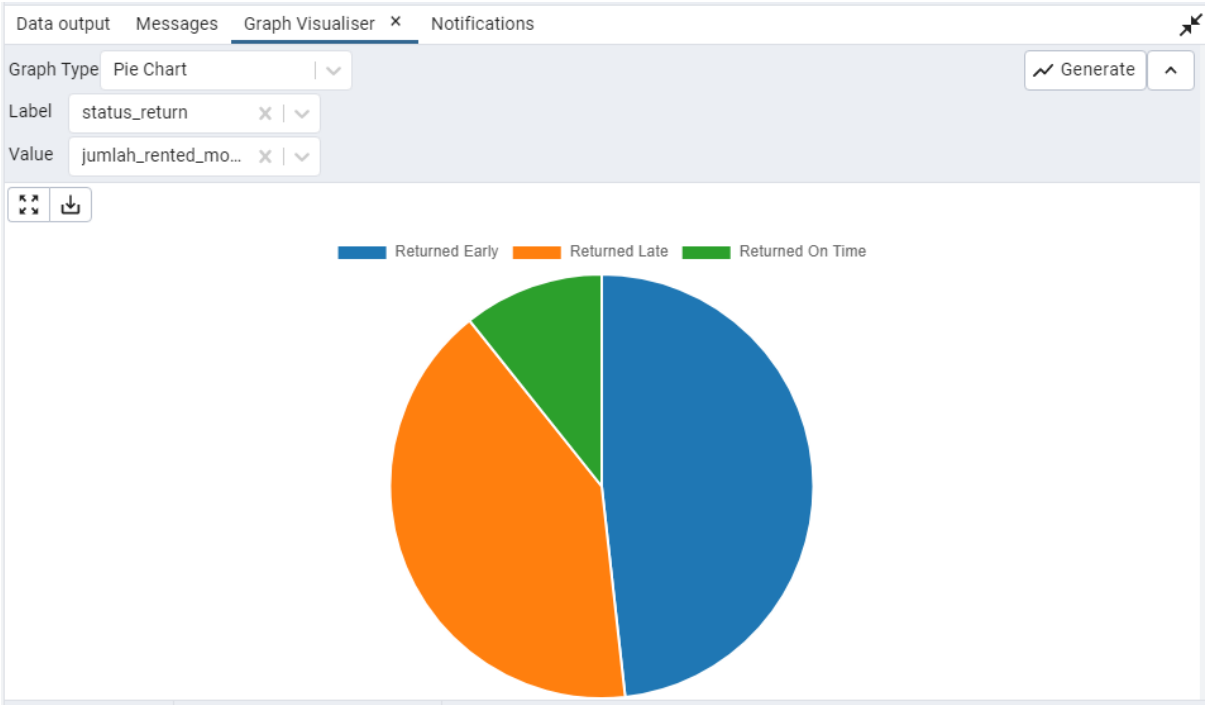
```
select case
when rental_duration > date_part('day', return_date - rental_date) then 'Returned Early'
when rental_duration = date_part('day', return_date - rental_date) then 'Returned On Time'
else 'Returned Late' end status_return,
count(*) jumlah_rented_movie
from public.film a
join inventory b on a.film_id=b.film_id
join rental c on b.inventory_id = c.inventory_id
group by status_return
```

   screenshot of query result:

Data output | Messages | Notifications

	status_return text	jumlah_rented_movie bigint
1	Returned Early	7738
2	Returned Late	6586
3	Returned On Time	1720

   result visualization:



5. What is the customer base in the countries where we have a presence?

sql query :

```
select a.country, count(d.customer_id) Jumlah_customers
from country a
join city b on a.country_id=b.country_id
join address c on b.city_id=c.city_id
join customer d on c.address_id=d.address_id
group by a.country
order by 2 desc
```

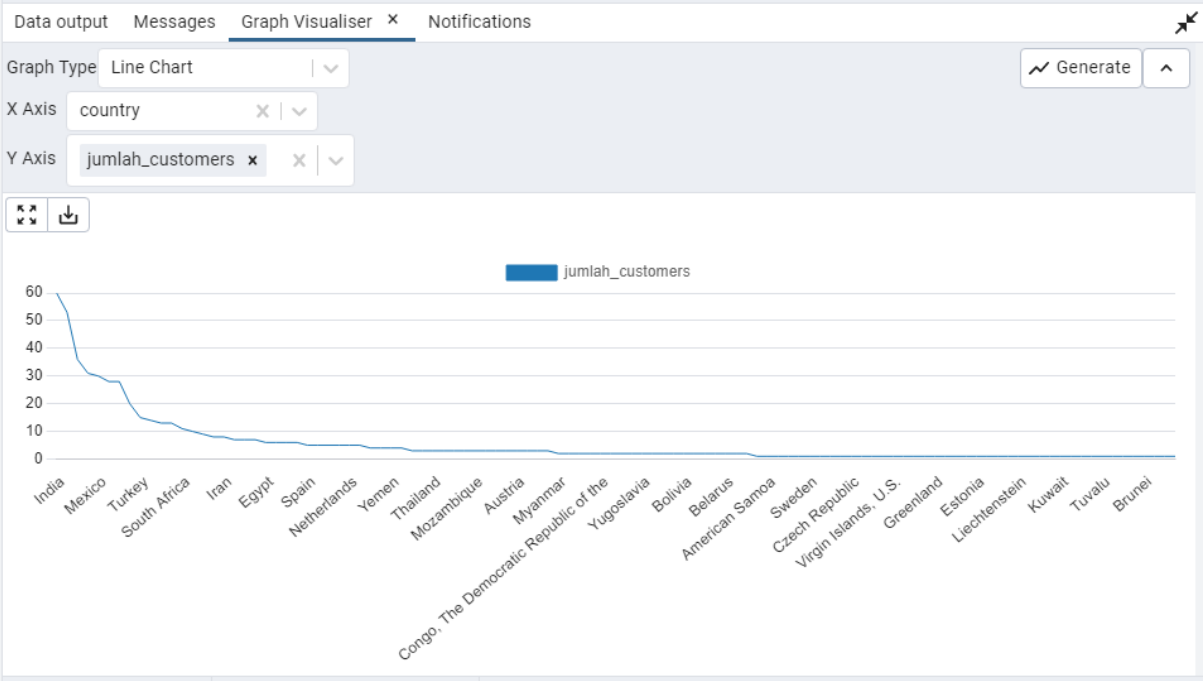
screenshot of query result:

Data output | Messages | Notifications

	country character varying (50)	jumlah_customers bigint
1	India	60
2	China	53
3	United States	36
4	Japan	31
5	Mexico	30
6	Brazil	28
7	Russian Federation	28
8	Philippines	20
9	Turkey	15
10	Indonesia	14
11	Argentina	13
12	Nigeria	13
13	South Africa	11
14	Taiwan	10
15	United Kingdom	9
16	Poland	8
17	Iran	8

Total rows: 108 of 108 | Query complete 00:00:09.548 | Ln 44, Col 1

result visualization:

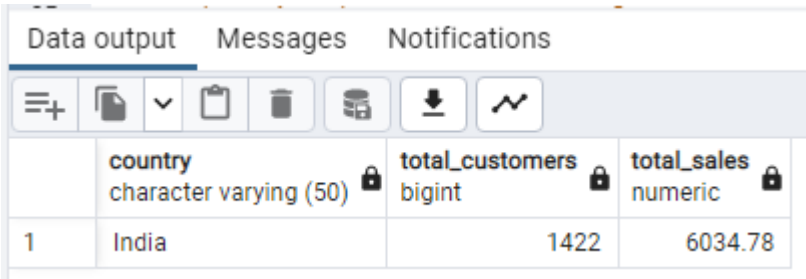


6. Which country is the most profitable for the business?

sql query:

```
select a.country, count(*) total_customers, sum(amount) total_sales
from country a
join city b on a.country_id=b.country_id
join address c on b.city_id=c.city_id
join customer d on c.address_id=d.address_id
join payment e on d.customer_id=e.customer_id
group by a.country
order by 3 desc
limit 1
```

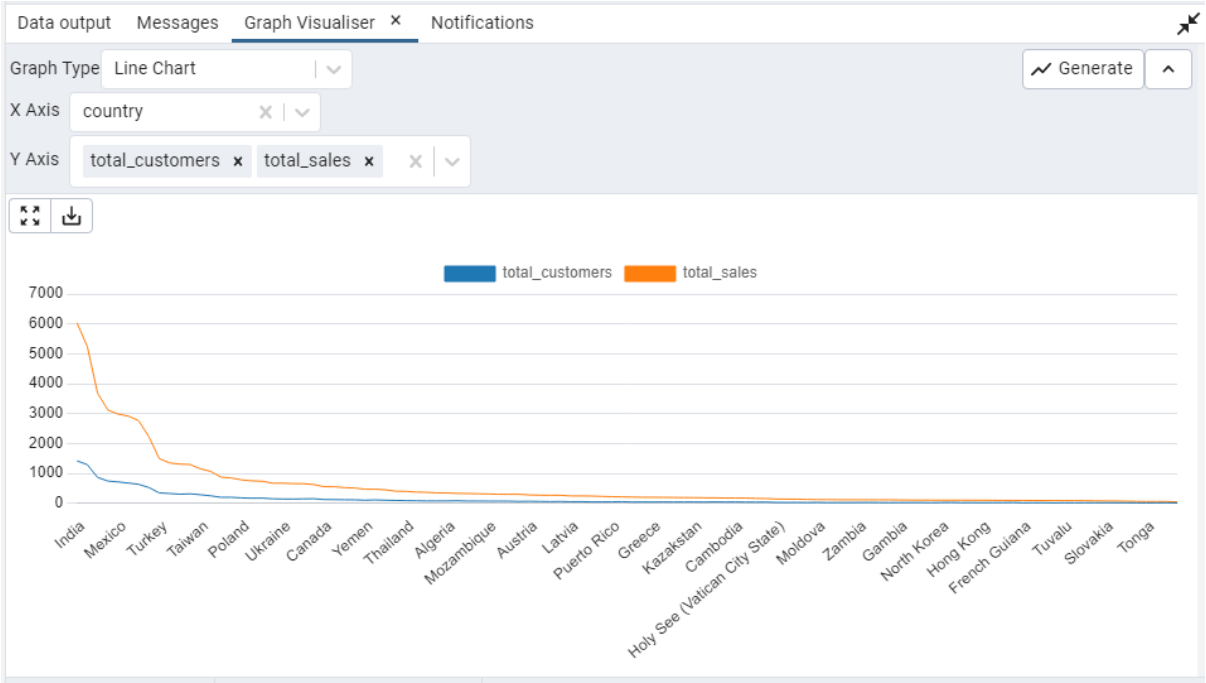
screenshot of query result:



Data output | Messages | Notifications

	country character varying (50)	total_customers bigint	total_sales numeric
1	India	1422	6034.78

result visualization (without limit):

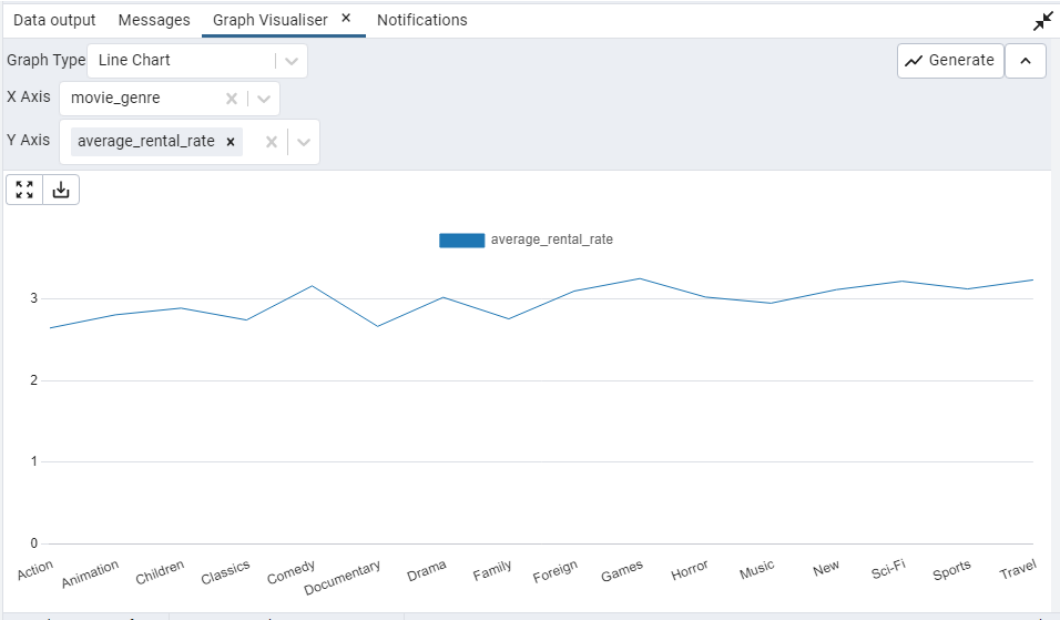


7. What is the average rental rate per movie genre (rating)?
sql query:

```
select a.name movie_genre, avg(rental_rate) average_rental_rate
from category a
join film_category b on a.category_id=b.category_id
join film c on b.film_id=c.film_id
group by a.name
order by a.name
```

screenshot of query result:

Data output Messages Notifications

	movie_genre character varying (25)	average_rental_rate numeric
1	Action	2.6462500000000000
2	Animation	2.8081818181818182
3	Children	2.8900000000000000
4	Classics	2.7443859649122807
5	Comedy	3.1624137931034483
6	Documentary	2.6664705882352941
7	Drama	3.0222580645161290
8	Family	2.7581159420289855
9	Foreign	3.0995890410958904
10	Games	3.2522950819672131
11	Horror	3.0257142857142857
12	Music	2.9507843137254902
13	New	3.1169841269841270
14	Sci-Fi	3.2195081967213115
15	Sports	3.1251351351351351
16	Travel	3.2356140350877193

result visualization (without limit):